

Evaluating Asian-specific pangenomes for HLA typing

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Abstract

Human leukocyte antigen (HLA) typing from short reads requires distinguishing closely related genes and resolving highly polymorphic alleles. Pangenomes provide complete haplotype sequences that can complement allele databases, but their usefulness depends on how those sequences enter the typing method. At BioHackathon Japan 2026, we constructed a 754-haplotype major histocompatibility complex (MHC) panel enriched for Asian and Arab donors and evaluated several ways to use it. In 40 East and South Asian donors, PanGenie with the full panel recovered more assembly-derived genotypes containing structural variation than an HPRC-only panel after we corrected a filter that discarded sites with missing training genotypes. The gains were 3.89 and 1.97 percentage points in the two strata. For named HLA genotypes, Locityper with the full panel matched 31 of 39 eligible experimental genotypes, compared with 28 for HPRC-only, 34 for T1K and 36 for SpecHLA. We also developed DōgoHLA, an extension of SpecHLA that combines panel-assisted read collection, repaired phasing and guarded reconstruction of graph-supported noncoding indels. In eight development donors, exact whole-gene reconstructions increased from 36 to 57 of 128 haplotypes. These experiments identify reference representation, callable-site retention and haplotype reconstruction as distinct determinants of HLA typing performance. Evaluation on new donors and independently constructed graphs is the next step.

Introduction

Human leukocyte antigen (HLA) genes encode molecules that present peptides to the immune system. HLA variation contributes to transplant compatibility and susceptibility to immune-mediated disease. Allele frequencies vary among populations, making population representation relevant to the design and evaluation of HLA reference resources (Gourraud et al., 2014). The major histocompatibility complex (MHC) also contains duplicated genes and structural variation. An HLA typing method must assign reads to the correct locus, distinguish the two inherited alleles and, for full-sequence typing, reconstruct each allele across coding and noncoding regions.

Existing methods address different parts of this problem. HLA*LA projects read alignments onto a population reference graph to infer HLA types (Dilthey et al., 2019). T1K estimates HLA and KIR allele abundances from reads aligned to allele references (Song et al., 2023). SpecHLA assigns reads to HLA loci, uses local assembly to improve alignment in divergent regions, and phases variants to reconstruct diploid gene sequences (S. Wang et al., 2023). Pangenome methods provide a complementary use of assembled haplotypes: PanGenie combines k-mer counts with a reference haplotype panel to genotype variants (Ebler et al., 2022), while Locityper selects pairs of locus haplotypes using read alignment and depth (Prodanov et al., 2025). These approaches make the choice of reference sequences and their representation part of the inference problem. A larger panel can add informative alleles while also changing which sites remain callable.

We investigated how an MHC panel enriched for Asian and Arab haplotypes can support HLA typing. First, we combined assemblies and graph-derived sequences from several pangenome projects, annotated HLA genes and constructed a whole-MHC graph. We then compared a

graph-consensus pipeline with direct-read typing and evaluated full and HPRC-only reference panels in 40 East and South Asian donors. We assessed graph-variant recovery, named HLA genotypes and RCCX/C4 structural signatures as separate outcomes. Finally, we developed DōgoHLA, a SpecHLA extension that uses the panel for read collection and selected structural reconstruction. This report presents the completed panel comparisons and eight-donor development results; the prospective DōgoHLA extension was ongoing at the analysis cutoff on 18 September 2026. The project repository is <https://github.com/leechuck/pangenome-bh26>; the appendix identifies the working analysis artifacts and their deposit status.

Methods

Haplotype panel and sequence annotation

We extended a 610-haplotype panel comprising the Arab Pangenome Reference (APR; 106 haplotypes) (Nassir et al., 2025), HPRC release 2 (464), JaSaPaGe Saudi (18), JaSaPaGe Japanese (20), GRCh38 and CHM13. We added 28 haplotypes from 14 K-PanRef individuals (Shin et al., 2026) and 116 from 58 Chinese Pangenome Consortium (CPC) individuals (Y. Wang et al., 2026). These additions used paths from the projects' Minigraph-Cactus graphs. We extracted K-PanRef haplotype paths with `vg paths -F`. For CPC, we used `odgi` to identify nodes in the CHM13 chromosome 6 interval 28–34 Mb and extracted each haplotype walk spanning those nodes. We retained segments with at least 50 kb of matching bases and mapping quality (MAPQ) ≥ 20 in a minimap2 `asm20` alignment to the GRCh38 MHC (Li, 2018). Three CPC samples already represented in HPRC were excluded from the additions.

The resulting panel contains 754 haplotype entries (Table 1). Excluding the two reference entries and five duplicate donor assembly pairs gives 742 haplotypes from 371 name-reconciled donors for donor-level analyses. Population metadata divide HPRC into Japanese (JPT), other East Asian (CHB, CHS, CDX, KHV and HG005), HG002 (Ashkenazi), and remaining samples. HG002 contributes two haplotypes from one individual. K-PanRef and CPC sequences inherit the clipping applied during construction of their source graphs; this can shorten private insertions.

Table 1: Haplotypes in the panel, grouped by source and population stratum.

Cohort / stratum	Haplotypes	Population	Source
APR	106	Arab (UAE)	Assembly
HPRC r2, Japanese	32	Japanese (1000G JPT)	Assembly
HPRC r2, HG002	2	Ashkenazi, one donor	Assembly
HPRC r2, other East Asian	70	CHB/CHS/CDX/KHV, HG005	Assembly
HPRC r2, remaining samples	360	Mixed	Assembly
JaSaPaGe, Saudi	18	Arab (Saudi Arabia)	Assembly
JaSaPaGe, Japanese	20	Japanese (1000G JPT)	Assembly
K-PanRef	28	Korean	Graph path
CPC	116	Chinese	Graph path
GRCh38 / CHM13	2	Reference	Reference
Total	754		

We used the Common Workflow Language workflow `hla_pangenome.cwl` to extract the extended MHC with `pgr-tk pgr-query` (Chin et al., 2023) and annotate gene structures and allele identities with `Immuannot v3` (Zhou et al., 2024). The extraction interval was GRCh38 chr6:28,510,120–33,480,577 with 100 kb flanks. Original annotations used IPD-IMGT/HLA v3.55, IPD-KIR v2.13 and RefSeq C4. We regenerated per-gene sequences, `pgr-tk` bundle decompositions, and `pgrb/odgi` graphs for the enlarged panel (Garrison et al., 2024; Guarracino et al., 2022). The subsequent coding-sequence audit used frozen IPD-IMGT/HLA v3.65.0

labels. Complete coding sequences (CDS) required valid strand and interval extraction and passed completeness checks; ambiguous or unmatched labels remained unresolved.

Whole-MHC graph and consensus-based typing

We built a whole-MHC graph from all 754 entries with Cactus v3.3.0 and the Minigraph-Cactus algorithm (Hickey et al., 2024), using GRCh38 and CHM13 as reference paths. The command requested clipped and full GFA/GBZ graphs, Giraffe indexes, VCF, full odgi output and visualisations. The build used 32 cores and 240 GB on the NIG supercomputer. We assessed mapping on MHC-recruited short reads from HG00096 using vg Giraffe v1.76.1 with eight threads (Sirén et al., 2021).

For an initial typing comparison, we aligned 1000 Genomes reads with Giraffe, called variants with `vg call`, phased them with WhatsHap (Patterson et al., 2015), and generated two haplotype consensus sequences with `bcftools consensus`. We annotated these sequences with Immunoanot using IPD-IMGT/HLA v3.55.0 and v3.65.0. Two separate T1K runs provided direct-read comparisons. The analysis used the completed outputs available in the archived snapshot of the 2,504-sample cohort; 955 cohort members overlap the experimental HLA resource of Gourraud et al. (Gourraud et al., 2014).

We compared unordered allele pairs at one-field and numeric two-field resolution. Pairwise comparisons covered A, B, C, DRA, DRB1, DQA1, DQB1, DPA1 and DPB1; comparisons with experimental typing covered A, B, C, DRB1 and DQB1. We removed Immunoanot's unresolved `:new` suffix before truncating names and included only calls with sufficient numeric fields in conditional concordance. Each method therefore has its own denominator in this initial analysis. For the assembly annotation benchmark, we retained the experimental ambiguity lists and required two eligible current-CDS labels per donor and locus.

Full versus HPRC-only panels in 40 donors

We selected 20 East Asian (EAS) and 20 South Asian (SAS) 1000 Genomes donors and assigned them to five population-balanced folds with known families kept together. For each fold, we excluded test donors, aliases and known relatives from both a full reference panel and an HPRC-only panel. The panels contained approximately 363 and 224–225 training donors, respectively. Both used the same whole-MHC graph, whose construction included the test assemblies. This design evaluates held-out haplotype inference conditional on a shared graph topology.

We decomposed the graph into top-level variant sites relative to GRCh38 with `vg deconstruct` and genotyped identical MHC-recruited reads with PanGenie v4.2.1. The initial panel builder removed any site with a missing training genotype. We revised this rule to retain known phased alleles and encode unknown alleles as missing, using PanGenie's native support for phased missingness. Alternate alleles entered a panel only when observed in its training haplotypes. Both panel arms used the revised rule and the original donor folds.

Variant truth comprised the test donors' assembly-derived allele sequences at graph sites within GRCh38 chr6:28,510,121–33,480,577 (1-based, inclusive). We excluded missing, conflicting and non-ACGT truth and required each allele's reference span to fit the interval. A correct genotype required an exact match of the unordered pair of allele sequences. Missing sites and no-calls counted as failures in recovery across all eligible truth genotypes. An SV-bearing genotype contained at least one allele whose length differed from the reference by ≥ 50 bp; success required the complete bubble allele pair, including embedded small variants. We also compared pure single-nucleotide variant (SNV) sites shared by both original graph outputs and the published NYGC GRCh38 PASS callset, keeping that comparison universe fixed after the repair. The NYGC arm used the October 2020 filtered, phased release from joint whole-genome calling.

To obtain named HLA genotypes, we used Locityper v1.7.4 (Prodanov et al., [2025](#)) on intact source locus sequences from the same training panels, with MHC-recruited reads and regional background calibration. Predictions received a numeric two-field name when their frozen exact-CDS labels resolved to one type. The assembly endpoint covered eight genes (A, B, C, DPA1, DPB1, DQA1, DQB1 and DRB1). Experimental labels supplied 39 eligible genotypes at five genes in eight EAS donors. T1K v1.0.6 and SpecHLA retained their frozen predictions and native databases; SpecHLA used IPD-IMGT/HLA v3.38.0, while T1K used the run's current-IPD reference. T1K required positive call quality and ambiguity that collapsed to a single numeric two-field allele. Unresolved predictions counted as failures. We estimated paired differences with 10,000 donor bootstrap resamples within each ancestry stratum, conditional on the fixed sites and folds. The random seed was derived from SHA256 of the analysis date (20260917), endpoint and baseline; intervals were unadjusted for multiple comparisons.

RCCX/C4 structural typing

We evaluated two short-read prototypes for the RCCX region, which contains C4A/C4B copy-number and long/short variation. The first fitted pairs of reference paths to sampled canonical 31-mer counts in 18 pilot and 88 additional donors. Markers required support in at least three training families, MHC locus specificity and base quality ≥ 20 ; overlapping mates contributed once per fragment. A correction for total-copy dosage was estimated on the 18 pilot donors and applied to the 88 additional donors. Calibrated ordinary reference depth supplied a simpler comparator. We scored coarse DRB gene-content signatures separately.

The second prototype used 299 targeted probes after development on the preceding 106 donors and was evaluated on 24 additional donors. Test families were excluded from probe discovery, training statistics and candidate paths. The probes measured C4 A/B and long/short marginal dosages; a reference-pair model then inferred complete structural signatures. Comparators were the earlier sketch, calibrated reference depth and C4Investigator (Marin et al., [2024](#)). C4Investigator's total-copy calibration used the 18 pilot donors. Structural truth comprised assembly annotations, including provisional CYP21/TNX prototype assignments.

DōgoHLA development and evaluation

DōgoHLA v0.1.0 extends SpecHLA v1.0.12 with panel-assisted read collection, corrected scoring of phase alternatives, IPD-IMGT/HLA v3.65 phase references and graph-supported reconstruction of selected noncoding indels. The phase repair groups scores for complementary haplotypes under the same phase alternative. For DRB1, we preserved reference paths in a normalised pggg graph, aligned reads with Giraffe and genotyped represented alleles. We decomposed confident homozygous graph alleles into primitive long indels with affine-gap alignment and applied eligible noncoding changes while preserving read-phased coding variants. Graph calls required PASS status, genotype and variant quality ≥ 20 , depth ≥ 10 and at least five reads supporting the alternate allele. Candidate indels were ≥ 50 bp and required valid reference sequence, mappable boundaries and unique flanks. We also removed the fixed 11-kb DRB1 query crop in the naming step. Frozen thresholds and code snapshots are retained with the experiments.

Development used eight fold-0 donors, with their families excluded from reference inputs. We compared reconstructed sequences with assembly-derived truth at eight genes per donor, assigning the two haplotypes to minimise total global edit distance. Exactness required a global whole-gene match; masked N bases counted as mismatches. We separately scored eligible two-field genotypes. Controls compared phasing changes, phase-reference databases, graph reconstruction and naming-only changes. The prospective extension froze the method before evaluation on the remaining 32 donors, with native SpecHLA and a DōgoHLA arm without graph reconstruction as comparators. This extension was incomplete at the report cutoff.

Results

MHC recovery and panel composition

We first assessed how much of the reference MHC interval was recovered from each input. Of 752 non-reference haplotypes, 747 reached ≥ 0.99 coverage and 720 were represented by a single extracted segment (Figure 1). Four CPC haplotypes and one JaSaPaGe Saudi haplotype had lower coverage, with minima of 0.92 and 0.96, respectively. These measurements quantify coverage of the GRCh38 interval. The source clipping inherited by CPC and K-PanRef also affects the representation of sequence outside that interval's alignment.

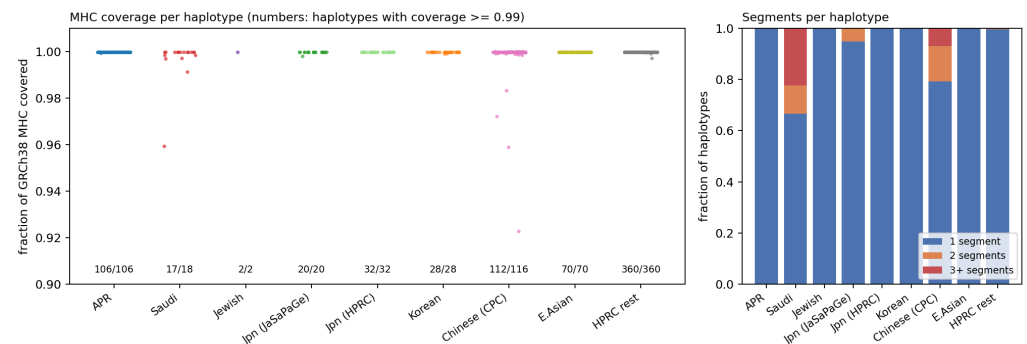


Figure 1: MHC extraction across 752 non-reference haplotypes. Left: coverage of the GRCh38 MHC interval, with counts at coverage ≥ 0.99 . Right: extracted segments per haplotype. CPC and JaSaPaGe Saudi contain the five lower-coverage haplotypes; CPC and K-PanRef are the graph-derived input cohorts.

Allele frequencies differed among the nine source/population strata (Figure 2). For example, HLA-A*11:01 occurred in 29% of CPC and 26% of other East Asian haplotypes, while A*24:02 occurred in 35–41% of the Japanese strata. These are frequencies among sampled panel haplotypes. At DRB1, 57–79% of annotated copies across strata lacked an exact full-length match to IPD-IMGT/HLA v3.55, motivating the subsequent current-database and CDS audits.

HLA allele landscape by cohort (HPRC r^2 split by population: HPRC Jewish = HG002 only) across 752 assembled haplotypes (Immunogen on assembly-derived MHC)



Figure 2: HLA annotations across nine strata. Top: HLA-A, HLA-B and HLA-DRB1 allele frequencies. Bottom: secondary DRB genes, C4A/C4B long/short forms and the fraction of copies lacking a full-length database match. The HPRC-Jewish column contains HG002 alone.

We used external 1000 Genomes SNP calls to distinguish near-homozygosity from duplicated assembly content (Figure 3). NA18976 and NA19909 carried fewer than 100 heterozygous SNPs per 100 kb across the MHC, supporting near-homozygosity. In contrast, NA18952 had heterozygous read-derived calls across the region but only ten substitutions between its two JaSaPaGe haplotypes. This supports duplication of one haplotype during assembly; the donor-level panel retained the HPRC assembly for this individual.

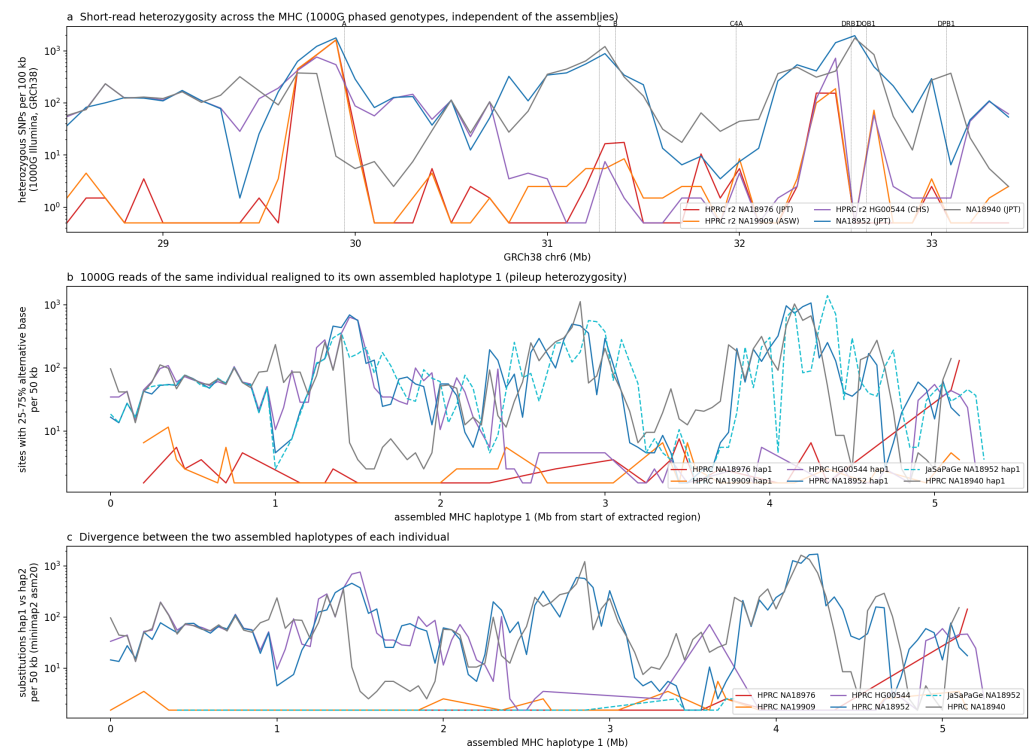


Figure 3: MHC heterozygosity from 1000 Genomes data compared with assembly haplotypes. Low external heterozygosity supports near-homozygosity in NA18976 and NA19909; discordance between external heterozygosity and the two JaSaPaGe sequences identifies NA18952 for assembly QC.

Coding-sequence audit and experimental HLA concordance

We checked assembly-derived HLA labels against current database sequences and experimental typing. HG02717's HLA-DQB1 sequence, previously labelled as a candidate new allele, matched registered DQB1*02:180. The NA20346 HLA-DPA1 candidate had local support from 24 of 51 reads in the initial analysis. The broader complete-CDS audit identified 47 distinct protein candidates absent from the frozen IPD-IMGT/HLA v3.65.0 database. Three candidates, NA18620 HLA-C and HG02976 and NA19159 HLA-DRB1, had at least five supporting fragments at each tested distinguishing SNP. These observations support the tested local bases; full-allele phasing and orthogonal sequence validation remain steps in candidate assessment.

Current exact-CDS labels agreed with the Gourraud experimental genotypes in 62/64 HLA-A, 64/64 HLA-B, 58/63 HLA-C, 59/63 HLA-DRB1 and 58/61 HLA-DQB1 comparisons. Ten of the 14 discrepancies had possible antigen-binding-exon compatibility under frozen G-group definitions. Among the remaining four, NA18943 A/DRB1 and NA19007 A had independent local read or cross-assembly support for the assembly bases; NA18608 DRB1 remained unresolved because several probes were non-unique. A complementary comparison between FuFiHLA (Hu et al., 2026) and Immuannot assessed agreement of annotations on the same assemblies (Figure 4). Together, these analyses establish the eligible sequence labels and identify discrepancies for further review.

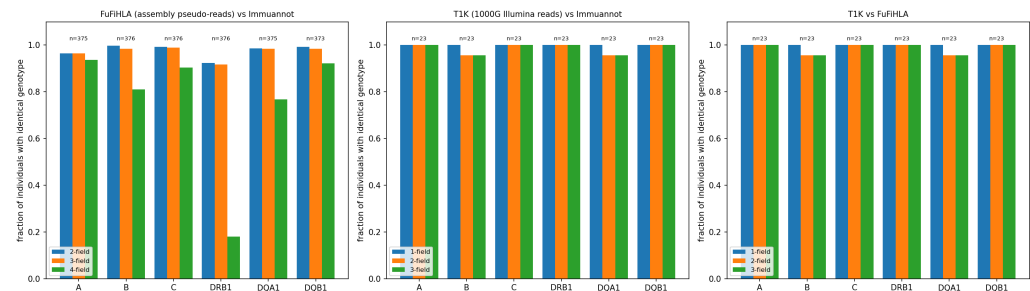


Figure 4: Agreement between FuFiHLA, Immuannot and T1K at one- to four-field resolution on overlapping assembled donors. Denominators depend on the method pair, locus and available resolution.

Graph-consensus typing resolves a limited subset of eligible genotypes

The whole-MHC graph contained 417,896 nodes and 579,137 edges, representing 5.89 Mb of sequence, and took 3 h 43 min to construct. In the HG00096 mapping test, 654,059 of 657,052 read pairs aligned (99.5%), including 625,590 at MAPQ ≥ 20 (95.2%), in 26 s. This established a working mapping resource for the typing experiments.

We then compared HLA calls from graph-derived consensus sequences with direct-read calls. Pairwise agreement varied with method, locus and resolution (Figure 5). The two database annotations of identical consensus sequences also differed, particularly at DRB1, showing that allele designation depends on the database as well as the reconstructed sequence. These comparisons motivate separate evaluation of sequence reconstruction and naming.

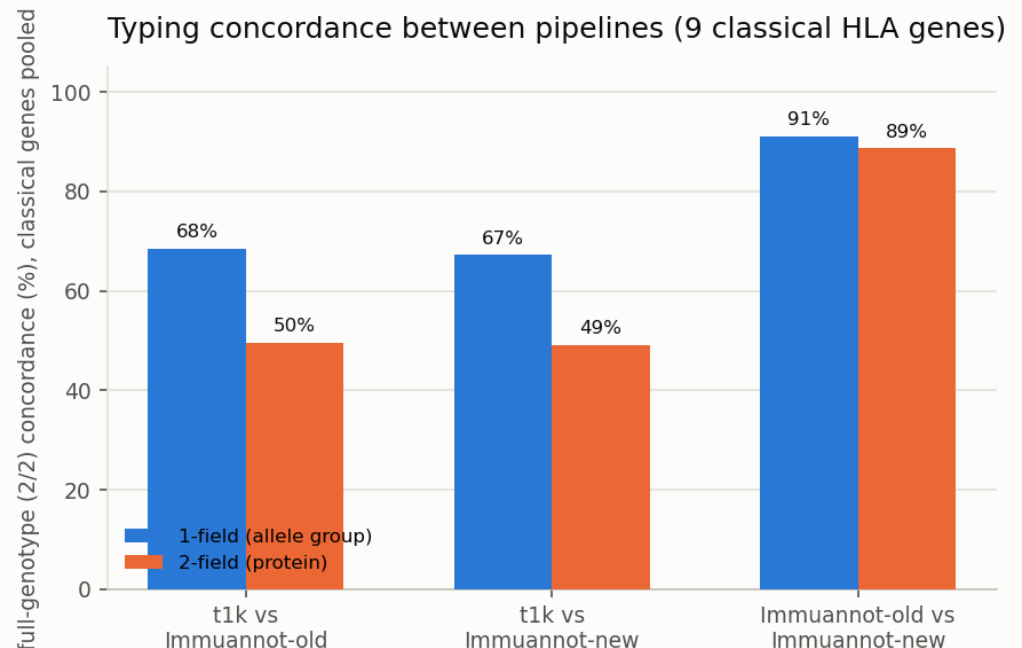


Figure 5: Pairwise HLA genotype agreement between two T1K runs and Immuannot annotations of graph-derived consensus sequences using two database releases. The plotted comparison pools nine HLA genes.

Against experimental typing, T1K had 98.8% genotype concordance among 1,843 checkable calls, the separate T1K run had 96.5% among 4,308, and each Immuannot database annotation

had 98.5% among 613 checkable calls (Figure 6). The archived analysis reports two-field resolution for approximately 90% of T1K's eligible calls and 33% of the graph-consensus calls. These conditional concordances refer to method-specific subsets. The practical finding is the larger number of resolved genotypes from T1K, which motivated evaluation of alternative uses of the haplotype panel.

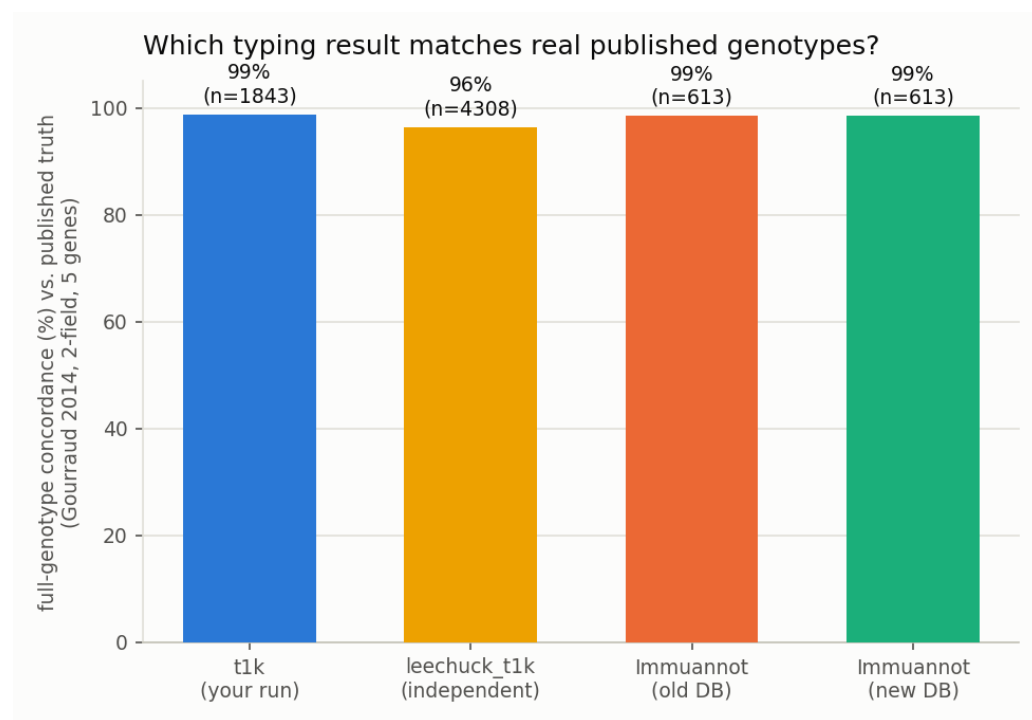


Figure 6: Conditional two-field concordance with published experimental typing at five genes. Each bar uses that method's resolved calls: 1,843 for the pipeline T1K run, 4,308 for the separate T1K run and 613 for each Immuannot annotation.

Retaining callable sites improves recovery with the full panel

The initial PanGenie analysis exposed a loss of callable sites when the panel builder required complete training genotypes. Allowing phased missing alleles restored 19,043–19,072 sites per fold in the full panel and 221–227 in HPRC-only. Every previously retained site remained, and every repaired HPRC-only site was present in the repaired full panel.

With missing sites and no-calls counted as failures, the repaired full panel recovered more SV-bearing assembly genotypes than repaired HPRC-only in both ancestry strata (Table 2). The paired gain was 3.89 percentage points in EAS (95% donor bootstrap interval 2.40–5.29) and 1.97 points in SAS (0.32–3.47). This supports the use of additional reference haplotypes for exact bubble-genotype recovery within the shared graph.

Table 2: Exact recovery of SV-bearing diploid bubble genotypes after callable-site repair.

Stratum	Eligible genotypes	HPRC-only	Full panel
EAS	1,465	1,060 (72.35%)	1,117 (76.25%)
SAS	1,577	1,188 (75.33%)	1,219 (77.30%)

On the fixed 1,007,636 shared SNV comparisons per stratum, concordance was 99.8217% for the full panel, 99.7749% for HPRC-only and 99.6907% for the published linear callset in

EAS; the corresponding SAS values were 99.8272%, 99.8039% and 99.6405%. The full-minus-HPRC paired intervals were 0.0148–0.0828 percentage points in EAS and –0.0060–0.0573 in SAS. These small shared-site differences complement the larger recovery gains from retaining previously omitted sites.

Named HLA performance depends on the reference and truth source

We evaluated whether the expanded panel also improved named HLA genotypes. An initial whole-locus PanGenie representation produced predominantly unresolved calls because its marker selection left few informative k-mers shared across complete alleles. Locityper provided a usable locus-haplotype inference method for the same panels. With Locityper, the full panel recovered more eligible HLA genotypes than HPRC-only on both assembly-derived and experimental labels (Table 3).

Table 3: Exact unordered numeric two-field genotypes. No-calls count as failures. Experimental truth covers five loci in eight EAS donors; assembly truth covers eight loci in 20 donors per stratum.

Method	Experimental EAS	Assembly EAS	Assembly SAS
Full panel + Locityper	31/39	137/159	143/160
HPRC-only + Locityper	28/39	134/159	139/160
T1K	34/39	134/159	120/160
SpecHLA	36/39	126/159	110/160

SpecHLA had the highest concordance on the experimental endpoint. The full panel's gain over HPRC-only was 7.69 percentage points, with a paired donor interval of –2.78 to 20.00. The assembly-derived endpoint favoured full-panel Locityper, with smaller gains over HPRC-only: 1.89 points in EAS (–1.87 to 5.66) and 2.50 in SAS (0.00 to 5.62). The truth-source audit found compatible historical and current assembly labels in 36 of 38 overlapping eligible genotypes, with two DRB1 disagreements. The differing method rankings therefore need to be interpreted with the endpoint and reference database specified.

C4 dosage is more readily recovered than complete structural pairs

The first structural prototype recovered coarse DRB gene content in all 106 donors. Its uncorrected RCCX path and binary-marker variants recovered 38/106 and 64/106 complete structural pairs, respectively. After pilot-only dosage calibration, the corrected method recovered total copy number in 88/88 additional donors and complete signature pairs in 73/88. Calibrated ordinary reference depth also recovered total copy number in 88/88.

In the subsequent 24-donor experiment, the targeted assay, calibrated reference depth and pilot-calibrated C4Investigator each recovered total C4 copy number in 24/24 donors. The targeted assay recovered 21/24 complete structural pairs, compared with 22/24 for the earlier sketch. These results distinguish accurate total dosage from the more difficult assignment of C4 forms to complete reference structures. The observed errors involved incorrect A/B dosage, ambiguous linkage between A/B and long/short forms, and a structure absent from the reference panel.

DōgoHLA improves whole-gene reconstruction in development donors

We next tested changes to phasing and sequence reconstruction within SpecHLA. Across the eight development donors, the selected DōgoHLA candidate increased exact whole-gene matches from 36/128 in the saved native SpecHLA runs to 57/128. Total global edit distance decreased from 97,965 to 75,634 (22.8%), and correct two-field genotypes increased from 61/63 to 62/63. Edit distance improved in all eight donors. These values use global whole-gene alignment throughout.

The controls localised the improvements. With panel-assisted read collection held fixed, complete phasing repair and updated phase references increased exact matches from 43/128 to 57/128. Adding panel sequences to the phase-reference database left exactness at 57/128. The final guarded graph-indel step reduced total edit distance from 76,924 to 75,634 while retaining 57/128 exact sequences and 62/63 correct genotypes. The reduction came from supported noncoding DRB1 insertions in two donors. An earlier whole-block replacement had overwritten coding differences and reduced two-field correctness to 60/63; decomposing supported changes into eligible noncoding indels preserved the read-phased coding alleles. The selected method therefore combines phasing repair with a narrower structural reconstruction step. Its prospective 32-donor extension will assess these choices with the method fixed before inference.

Discussion

We constructed an MHC panel enriched for Asian and Arab haplotypes and evaluated how its sequences contribute to variant genotyping, named HLA typing and gene reconstruction. The strongest evidence for the full panel came from recovery of SV-bearing assembly genotypes after callable-site repair. This result connects reference diversity to an implementation requirement: added haplotypes are useful only when the panel representation preserves the sites at which they can contribute evidence. The smaller and less certain named-HLA gains also show why variant recovery and allele designation require separate evaluation.

The comparisons build on established graph, haplotype and sequence-reconstruction methods. PanGenie supplied variant inference, Locityper supplied complete-locus selection, and SpecHLA supplied read-based reconstruction. DōgoHLA extends the last approach with panel-assisted read collection, repaired phasing and selected graph-supported noncoding changes. In the development controls, phasing repair accounted for the increase in exact whole-gene matches; the guarded graph step reduced residual sequence errors in two DRB1 donors. A complete paired evaluation of DōgoHLA and its graph ablation will quantify the contribution of that step on additional donors.

The 40-donor panel comparison is conditional on a graph built with the test assemblies. The full panel is also larger than HPRC-only, so its observed effect combines panel size and composition. A size-matched augmentation experiment using graphs constructed exclusively from training donors would separate these effects. Experimental HLA labels are available for only eight EAS donors in this benchmark; additional experimental typing in SAS and Arab donors would extend the ancestry coverage. Harmonised allele databases and read recruitment would also make comparisons among typing methods easier to interpret. The published linear arm provides a shared-SNV comparison under its original joint-calling and filtering design.

Sequence quality and truth definition impose further constraints. Graph-derived inputs inherit clipping, some assembly haplotypes require QC, and historical HLA labels can resolve a different sequence region from a complete-CDS annotation. The candidate catalogue therefore retains local read evidence separately from complete-allele validation. Similarly, C4 dosage and reference-signature imputation describe different levels of structural information. Long reads spanning diagnostic sites and module boundaries would provide an independent test of the inferred RCCX arrangements. These follow-up experiments can establish when the expanded panel improves inference beyond the fixed-panel comparisons reported here.

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Appendices

Analysis provenance and reproduction

The companion repository (<https://github.com/leechuck/pangenome-bh26>) contains the extraction and annotation workflow (hla/), graph construction instructions (hla/docs/MHC_GRAPH_README.md), graph-consensus comparison scripts and tables (hla/analysis/gc_*.py, hla/results/tables/gc_*.tsv), and assembly annotation audit (hla-analysis/FINAL_REPORT.md, hla-analysis/REPRODUCIBILITY.md). In the working analysis repository, the completed 40-donor comparisons are documented in hla-asian50/refined/REPORT.md and hla-asian50/refined/README.md; the original analysis is retained in hla-asian50/accuracy/REPORT.md. Structural protocols, measurements and calibration are in hla-structural/ and hla-targeted/.

DōgoHLA development results and frozen code snapshots are documented in hla-speechla-pg/RESULTS-2026-09-18.md and hla-speechla-pg/IMPLEMENTATION.md. The prospective extension is specified in hla-speechla-pg/validation/20260918/PROTOCOL.md. The analysis directories retain donor manifests, exclusions, parameter settings, per-donor scores and machine-readable input/output checksums. Large sequence inputs and native outputs require restoration from the locations described in those instructions. The results in this report use completed analyses available at the 18 September 2026 cutoff; interim results from the ongoing DōgoHLA extension are excluded from the reported comparisons. Public deposit of the refined 40-donor report and DōgoHLA artifacts was pending at the availability check for this draft; their paths above identify local working artifacts.